

30.2-km D-Band Photonics-Assisted Wireless Reception Enabled by FPGA-Based Equalizer-Guided SFO Compensation

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Abstract An FPGA-deployed receiver is demonstrated for a 30.2-km photonics-assisted D-band wireless link with equalizer-guided SFO estimation and Farrow-based compensation. Using experimentally captured waveform data loaded into on-board RAM, the FPGA receiver successfully achieves 16-Gbaud QPSK transmission below the 20% SD-FEC threshold. ©2026 The Author(s)

Introduction

D-band wireless transmission has attracted increasing interest for beyond-5G/6G high-capacity links because of its large available bandwidth and the rapid progress of photonics-enabled terahertz systems [1–3]. Among the reported approaches, photonics-assisted schemes are particularly promising for high-frequency carrier generation and long-distance delivery. However, such demonstrations are still mainly verified by offline digital signal processing (DSP), whereas implementation-oriented FPGA-deployed photonics-aided receivers remain much less explored [4–6]. This gap between offline verification and practical real-time deployment becomes more pronounced in long-reach D-band links, where synchronization robustness and hardware efficiency must be jointly considered.

For hardware-oriented receivers, sampling-frequency offset (SFO) between independently clocked transmitter and receiver becomes a critical impairment, since it causes accumulated timing drift during long-duration operation. This drift degrades the sampling accuracy and further affects subsequent equalization and carrier recovery. In optical receivers, low-complexity and hardware-efficient timing/sampling-offset recovery has been extensively investigated, highlighting the importance of explicit SFO/clock-drift compensation for implementation-oriented DSP [7–10]. In this work, we develop an FPGA-deployed SFO-aware receiver DSP for the 30.2-km photonics-assisted D-band wireless link. The residual SFO-induced timing drift is estimated from the migration period of the dominant tap in a T/2-spaced adaptive equalizer and compensated by a Farrow-based variable fractional-delay interpolator. The received waveform is captured by a high-speed oscilloscope, quantized, and loaded into on-board RAM, after which the receiver DSP is executed on FPGA. This architecture preserves the original transmission setup while demonstrating on-board deployment

ment of hardware-friendly timing correction for long-distance D-band wireless reception.

Tab. 1: State-of-the-art photonic-assisted THz transmission works.

Ref.	Bit Rate (Gbps)	Wireless Distance (km)	Bit Distance Product (Gbps·km)	Offline /On-board
[11]	55.8	4.6	257.0	Offline
[12]	41.7	4.6	191.8	Offline
[13]	78.13	4.6	359.4	Offline
[14]	200.0	4.6	920.0	Offline
[15]	18.0	30.2	543.6	Offline
[16]	84.0	1.26	105.8	On-board
[17]	50.0	1.0	50.0	On-board
Our work	32	30.2	966.4	On-board

Equalizer-Guided SFO Compensation

Sampling-frequency offset (SFO), defined as $\varepsilon = (f_{rx} - f_{tx})/f_{tx}$, causes progressive sampling-phase drift in a clock-asynchronous receiver. The received signal before timing correction is

$$r[n] = x(nT_0 + \tau_0 + n\varepsilon T_0) + v[n], \quad (1)$$

where T_0 is the ideal sampling interval, τ_0 is the initial timing offset, $v[n]$ is noise, and $x(t)$ denotes the effective received waveform.

Under residual SFO, the optimal sampling phase drifts slowly over time. After digital down-conversion, a T/2-spaced adaptive equalizer is used to monitor this drift through the migration of its dominant tap. Let

$$k_p[q] = \arg \max_k |w_k[q]| \quad (2)$$

denote the dominant-tap index and let k_c denote the center tap. In a T/2-spaced equalizer, a transition of the dominant tap from k_c to $k_c \pm 2$ indicates that the accumulated timing drift has reached the neighboring symbol-spaced position. Therefore, if N_m timing-update intervals are required for this transition, the average fractional-delay increment per interval can be estimated as

$$\Delta\mu_q = \frac{s_q}{N_m}, \quad s_q = \text{sgn}(k_p[q] - k_c), \quad (3)$$

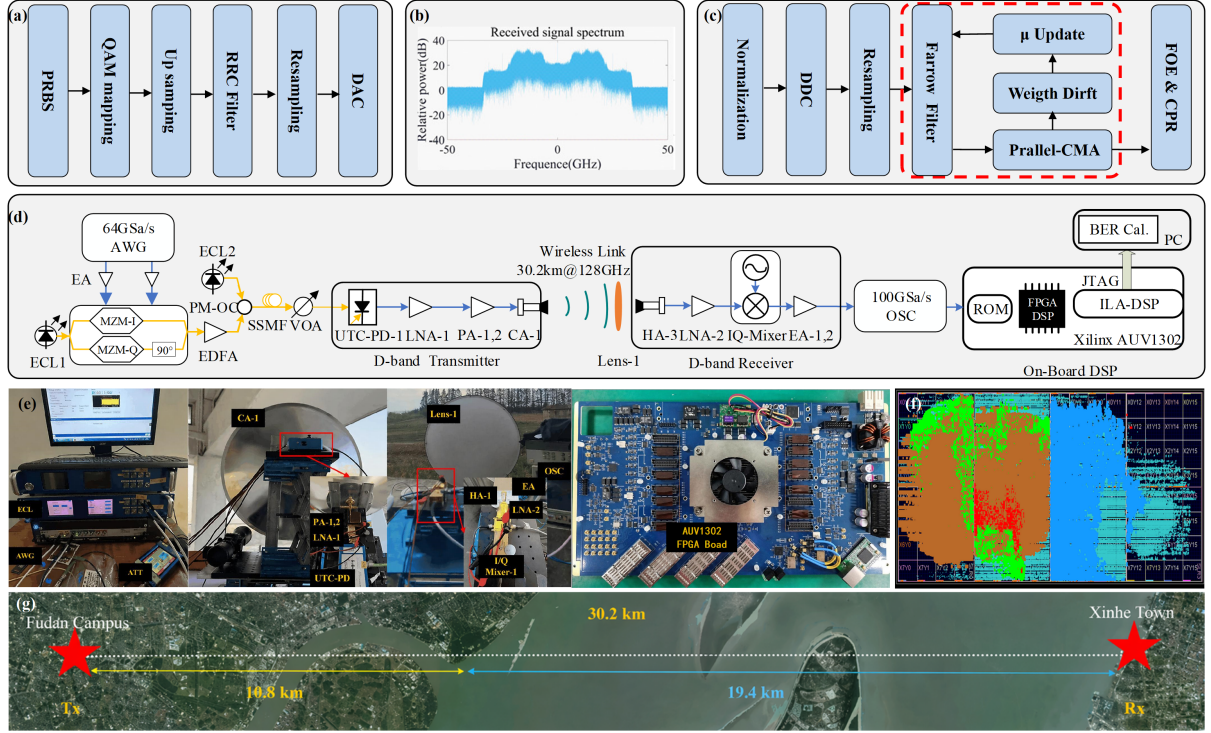


Fig. 1: Experimental setup and signal processing flow of the proposed wireless transmission system. (a) Transmitter offline DSP. (b) Spectrum of the received signal. (c) Receiver on-board DSP implemented on FPGA. (d) Schematic diagram of the full experimental setup. (e) Photographs of the outdoor field trial. (f) FPGA board netlist. (g) Geographic locations of the transmitter and receiver for the field experiment.

where s_q indicates the drift direction.

The estimated increment is then fed back to a Farrow-based fractional-delay interpolator, and the control parameter is updated by $\mu_{q+1} = \text{mod}(\mu_q + \Delta\mu_q, 1)$. In this way, the proposed receiver estimates residual SFO from the migration period of the dominant tap and iteratively corrects the accumulated timing error with low implementation complexity. The overall equalizer-guided SFO estimation and compensation procedure is summarized in Algorithm 1.

Algorithm 1 Equalizer-guided SFO estimation and timing correction

- 1: **Input:** received samples $r[q]$
- 2: **Initialize:** $\mu \leftarrow 0$, $N_m \leftarrow 0$, k_c
- 3: **for** each timing-update interval **do**
- 4: Update the T/2-spaced equalizer and obtain $k_p = \arg \max_k |w_k|$
- 5: $N_m \leftarrow N_m + 1$
- 6: **if** $k_p \geq k_c + 2$ **or** $k_p \leq k_c - 2$ **then**
- 7: Estimate the average fractional-delay increment $\Delta\mu \leftarrow \text{sgn}(k_p - k_c)/N_m$
- 8: Update the Farrow control parameter $\mu \leftarrow \text{mod}(\mu + \Delta\mu, 1)$
- 9: Reset $N_m \leftarrow 0$
- 10: **end if**
- 11: Perform Farrow-based retiming on $r[q]$ with control parameter μ
- 12: **end for**
- 13: **Output:** retimed samples

Experiment Setup

Fig. 1 summarizes the 30.2-km photonics-assisted D-band wireless experiment and the corresponding receiver DSP flow. A 16-Gbaud QPSK waveform generated by offline DSP was loaded into a 64-GSa/s AWG, and the photonics-assisted D-band signal was transmitted over a 30.2-km land/sea mixed path at 128 GHz. The received spectrum is shown in Fig. 1(b).

At the receiver, the signal was collected by a PTFE lens and horn antenna, amplified, down-converted by an IQ subharmonic mixer driven by a 143-GHz local oscillator, and captured by a 100-GSa/s oscilloscope. The quantized IF data were written into on-board RAM and processed on an AUV1302 board with a XCVU13P-2FLGA25771 FPGA. Operating at 250 MHz with 64-way parallelism, the platform supports 16-Gbaud signal processing. Figs. 1(e) and 1(g) show the field-trial photographs and the transmitter/receiver locations.

The implementation-oriented receiver DSP is shown in Fig. 1(c). Compared with the baseline offline DSP chain, the proposed equalizer-guided SFO estimation and Farrow-based compensation stage is inserted before the main equalization and carrier-recovery modules and deployed on FPGA. A T/2-spaced CMA equalizer monitors dominant-tap migration, and the corresponding migration interval is converted into an average fractional-delay increment for Farrow-based timing correction.

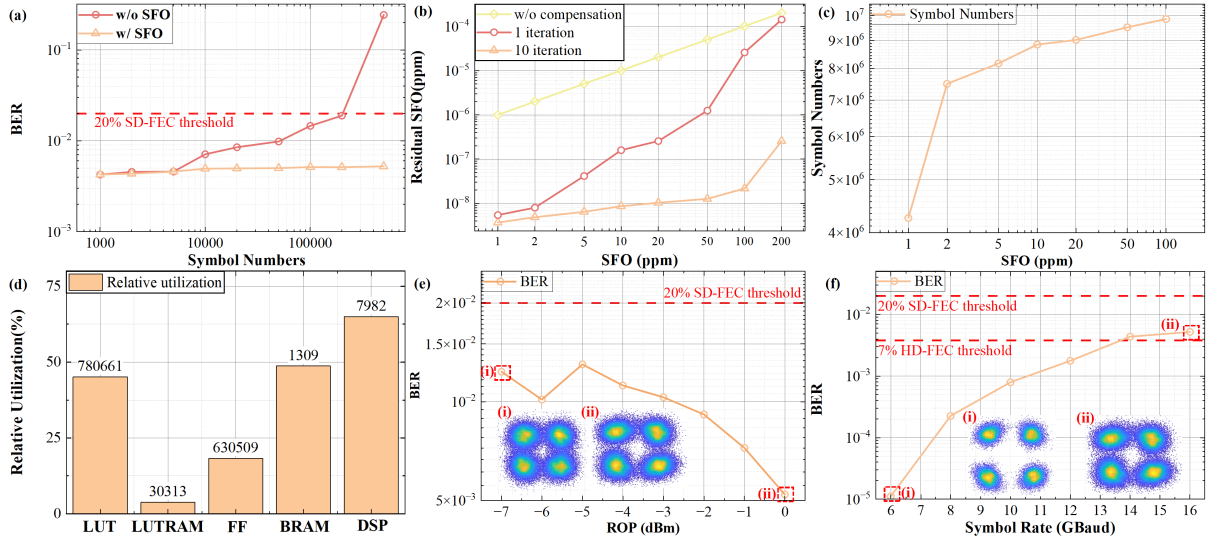


Fig. 2: Experimental results. (a) BER versus data length. (b) Residual SFO versus SFO. (c) Number of symbols required for startup acquisition versus SFO. (d) FPGA resource utilization. (e) BER versus ROP. (f) BER versus symbol rate.

After SFO compensation, the corrected samples are processed by branch combining, adaptive equalization, frequency-offset estimation, carrier-phase recovery, and decision-directed refinement. The FPGA hardware platform is shown in Fig. 1(f). Thus, Fig. 1 jointly presents the transmission setup, field-trial scenario, and on-board receiver DSP architecture.

Results

The transmission performance versus data length is first evaluated in Fig. 2(a). Without SFO compensation, the BER degrades rapidly as the data length increases because of the accumulated timing drift. In contrast, with SFO compensation, the BER remains much more stable over the whole tested range, verifying the effectiveness of the proposed timing correction scheme for long-duration reception.

The convergence behavior of the proposed equalizer-guided method is further evaluated in Fig. 2(b) and (c). As shown in Fig. 2(b), the residual SFO increases with the imposed SFO when no compensation is applied. After iterative compensation, the residual SFO is significantly reduced, and the suppression improves as the number of iterations increases, with ten iterations yielding the lowest residual error over the whole tested range. Fig 2(c) shows the number of symbols required for startup acquisition, i.e., the number of received symbols needed before the dominant-tap migration provides the first reliable update of the average fractional-delay increment. Under the present implementation settings, this startup overhead increases with SFO, indicating that a larger sampling-frequency mismatch requires more symbols to stabilize the equalizer-guided timing loop during initialization.

The FPGA implementation results are pre-

sented in Fig. 2(d), which shows resource utilizations. These results confirm that the complete receiver DSP, including the proposed SFO estimation and Farrow interpolation modules, was successfully deployed on the target FPGA platform with acceptable hardware overhead.

Figs 2(e) and (f) further show the BER performance under different received optical powers and symbol rates. In Fig. 2(e), the BER gradually decreases with increasing received optical power, and the inset constellation shows clear QPSK symbol clustering after receiver processing. In Fig. 2(f), the BER increases with symbol rate because the available system margin becomes smaller at higher transmission speeds. Overall, the measured BER remains below the 20% SD-FEC threshold, and 16-Gbaud QPSK transmission is achieved over the 30.2-km link, demonstrating the feasibility of the proposed FPGA-deployed receiver for long-reach photonics-assisted D-band wireless reception.

Conclusion

We demonstrated an FPGA-deployed equalizer-guided SFO estimation and Farrow-based compensation scheme for a 30.2-km, 128-GHz photonics-assisted D-band wireless receiver. The received waveform was loaded into on-board RAM, and the receiver DSP was executed on FPGA. By tracking the migration period of the dominant tap in a $T/2$ -spaced adaptive equalizer and correcting the accumulated timing error with a Farrow interpolator, the proposed method provides a hardware-efficient solution for long-duration clock-asynchronous reception. Experimental results show that the BER meets the 20% SD-FEC threshold for 16-Gbaud QPSK transmission, verifying the feasibility of the proposed architecture for practical long-reach D-band wireless receivers.

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